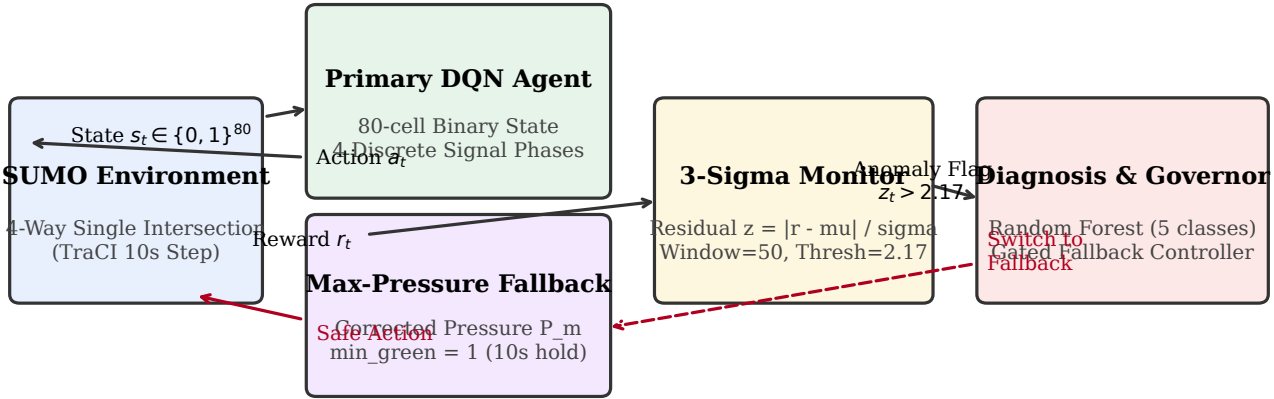
Primary DQN Agent
80-cell Binary State
4 Discrete Signal Phases
State $s_t \in \{0,1\}^{80}$
Action a_t
SUMO Environment
4-Way Single Intersection
(TraCI 10s Step)
Reward r_t
3-Sigma Monitor
Residual z = |r - mu| / sigma
Window=50, Thresh=2.17
Anomaly Flag
$z_t > 2.17$
Diagnosis & Governor
Random Forest (5 classes)
Gated Fallback Controller
Switch to Fallback
Max-Pressure Fallback
Corrected Pressure P_m
min_green = 1 (10s hold)
Safe Action